Global Database and Technical Analysis of Weft Yarn Materials: Engineering Properties and Fiber Morphology

The systematic classification and technical characterization of textile fibers are essential for the engineering of high-performance woven structures. In the context of weft yarn selection, materials must be evaluated based on their ability to withstand the unique mechanical stresses of the filling process—such as high-speed insertion, sudden tension changes, and abrasion against warp threads—while providing the desired aesthetic and functional properties in the final substrate. The following database serves as a comprehensive repository for textile engineering professionals, categorizing a vast array of natural, regenerated, synthetic, and industrial materials.

Weft Yarn Material Database

The following data structure provides a foundational catalog of the materials discussed in this report, including primary categories and specialized sub-types used in global textile manufacturing.

JSON

Natural Cellulosic Fiber Variants: Morphological and Technical Diversity

Natural cellulosic fibers are fundamentally composed of β-D-glucopyranose units linked by 1,4-glycosidic bonds. Their performance in weft applications is dictated by the degree of polymerization (DP), which can range from 2,000 to 5,000 in cotton and up to 10,000 in certain bast fibers.

Seed-Hair Fibers: The Cotton Hierarchy

Cotton remains the most utilized natural fiber for weft yarns due to its comfort, absorbency, and economic viability. However, significant technical variations exist between its three primary species: *Gossypium hirsutum*(Upland), *Gossypium barbadense* (Pima/Egyptian), and the rare *Gossypium herbaceum*.

Upland cotton accounts for approximately 90% of global production. In weft yarns, its ribbon-like convolutions—indentations in the fiber structure—provide the necessary cohesiveness for spinning. In contrast, extra-long staple (ELS) varieties like Pima and Egyptian cotton offer fiber lengths exceeding 32 mm, which reduces the number of fiber ends in the yarn cross-section, thereby increasing tensile strength and reducing pilling.

The moisture regain of cotton, standardized at 8.5%, is a defining characteristic. A critical phenomenon for weft processing is the increase in tensile strength when cotton is wet—approximately 20% higher than in the dry state. This is attributed to the presence of hydrogen bonds that reorganize in the presence of water molecules, increasing the molecular orientation within the amorphous regions.

Technical Properties of Seed-Hair Fibers

Fiber Variant	Tenacity (cN/tex)	Elongation %	Shrinkage %	Regain %	Thermal Behavior
Upland Cotton	22 – 26	5.0 – 10.0	3 – 5	8.5	Scorch at 218°C
Pima Cotton	28 – 32	6.0 – 9.0	2 – 3	8.5	High heat resistance
Egyptian Cotton	26 – 30	6.0 – 8.0	2 – 4	8.5	High heat resistance
Organic Cotton	22 – 28	5.0 – 10.0	3 – 6	8.5	Consistent with Upland
Kapok	12 – 15	1.8 – 4.2	1 – 2	10.0	Highly flammable

Bast and Leaf Fibers: Structural Reinforcements

Bast fibers—extracted from the phloem of plant stems—include flax, hemp, jute, and ramie. These fibers are characterized by high cellulose content (60-80%) and significant lignin (up to 13%), which provides structural rigidity and high tensile strength but limits elasticity. Flax (linen) is traditionally used in weft yarns for its "cool" hand-feel, a result of its high thermal conductivity compared to other natural fibers.

Jute, the "golden fiber," is the second most produced vegetable fiber after cotton. Its high lignin content makes it brittle and sensitive to moisture-induced degradation, which limits its use in fine apparel but makes it ideal for industrial sacking and geotextile wefts. Ramie is distinguished by its extreme tenacity and natural luster, though its stiffness requires careful decortication to avoid fiber breakage during weaving.

Hard leaf fibers, such as abaca and sisal, represent the pinnacle of natural tensile strength. Abaca, also known as Manila hemp, is the strongest known natural fiber, offering exceptional resistance to saltwater decomposition. In weft applications, these fibers are predominantly used for heavy-duty industrial fabrics, cordage, and specialty paper pulps.

Technical Properties of Bast and Leaf Fibers

Fiber	Tenacity (cN/tex)	Elongation %	Regain %	Density (g/cm³)	Specific Properties
Flax (Linen)	50 – 65	2.7 – 3.5	10 – 12	1.54	High thermal conductivity
Hemp	55 – 90	1.6	12.0	1.48	UV and alkali resistant
Jute	30 – 34	2.0 – 8.2	13.8	1.44	Brittle, photo-sensitive
Ramie	60 – 75	3.0 – 4.0	11.0	1.55	Silky luster, microbial
Abaca	90 – 120	1.1	10.5	1.50	Saltwater durable
Sisal	50 – 60	3.0 – 4.5	13.0	1.33	High friction coefficient
Coir	10 – 15	30.0	10.5	1.40	High extensibility

Natural Protein Fibers: Keratin and Fibroin Systems

Protein fibers are derived from animal sources and are primarily composed of alpha-keratin (hair/wool) or fibroin and sericin (silk). These materials are characterized by their high elasticity and moisture regain, though they are more sensitive to thermal and chemical degradation than cellulosic fibers.

Silk: The Only Natural Filament

Silk is a continuous protein filament secreted by silkworms. The most common variety, Mulberry silk (*Bombyx mori*), is produced in a controlled environment, resulting in a fine, uniform, and highly lustrous fiber. The cross-section of silk is triangular with rounded corners, which reflects light to produce its characteristic sheen.

Wild silks, such as Tussah, Eri, and Muga, provide different technical advantages. Muga silk, unique to the Assam region of India, is naturally golden and increases in luster with age. Eri silk, often called "peace silk" because the moth is allowed to emerge from the cocoon, has a shorter fiber length and a wool-like texture, making it highly effective for thermal insulation in weft yarns.

Silk Type	Tenacity (g/d)	Elongation %	Regain %	Elasticity	Dye Affinity
Mulberry	3.5 – 5.0	15 – 25	11.0	Medium	Acid/Basic Dyes
Tussah	3.0 – 4.0	20 – 30	11.0	Low-Medium	Good (Wild texture)
Eri	2.5 – 3.5	20 – 30	11.0	High	Good (Matte finish)
Muga	4.0 – 5.0	20 – 30	11.0	High	Excellent (Golden)

Wool and Keratinous Hair

Wool is distinguished by its complex morphological structure, including surface scales (cuticles) and a crimped configuration. These scales provide a "serrated" edge that entangles fibers during spinning, enhancing yarn strength. Wool is the most hygroscopic of all textile fibers, capable of absorbing up to 30% of its weight in moisture without feeling damp.

The resilience of wool—its ability to return to its original shape after deformation—is exceptionally high. In weft yarns, this property contributes to the crease resistance of fabrics. However, wool is a weak fiber, especially when wet, where its tenacity can drop by up to 20%.

Animal Fiber	Micron (μ)	Tenacity (g/d)	Regain %	Elasticity	Thermal Limit
Merino Wool	15 – 24	1.0 – 1.7	13 – 17	Excellent	Scorch 130 – 140°C
Cashmere	12 – 16	1.0 – 1.5	13 – 15	Good	Sensitive to heat
Mohair	24 – 40	1.5 – 2.0	13.0	Medium	Resilient
Alpaca	18 – 30	1.4 – 1.8	10 – 12	Good	Stable
Vicuna	10 – 12	0.8 – 1.2	13.0	Fair	Very heat sensitive

Regenerated Fibers: Advanced Polymer Engineering

Regenerated fibers involve the chemical dissolution of natural polymers followed by regeneration in fiber form. This allows for the manipulation of fiber cross-sections and degree of crystallinity, resulting in materials that combine natural comfort with synthetic uniformity.

Cellulose-Based Regeneration: Viscose to Lyocell

Viscose rayon is the most common regenerated cellulose fiber. While highly absorbent and soft, its mechanical performance is hampered by a significant loss of strength (up to 50%) when wet. This led to the development of Modal and Lyocell.

Lyocell (Tencel) represents the state-of-the-art in regenerated cellulose. Produced using an organic solvent (NMMO) in a closed-loop system, it offers higher tenacity than cotton in both wet and dry states. Modal fibers, such as those produced by Lenzing, are designed for high-wet-modulus (HWM), providing superior dimensional stability and shrinkage resistance compared to standard viscose.

Acetate and Triacetate Esters

Unlike viscose, acetate and triacetate are chemical esters of cellulose. Acetate has low abrasion resistance and a low melting point, but it provides a luxurious silk-like drape. Triacetate offers improved thermal stability, allowing for the creation of durable heat-set pleats in woven fabrics.

Regenerated Fiber	Tenacity (cN/tex)	Elongation %	Shrinkage %	Regain %	Dye Affinity
Viscose Rayon	22 – 26	15 – 30	High	11 – 16	High (Direct/Reactive)
Modal	30 – 35	15 – 20	Low	12 – 14	Excellent
Micro-modal	30 – 35	15 – 20	Very Low	12 – 14	Excellent
Lyocell (Tencel)	38 – 44	14 – 18	Minimal	11.5	High
Acetate	11 – 13	25 – 45	Moderate	6.0	Disperse Dyes
Triacetate	11 – 13	25 – 35	Low	3.5	Disperse Dyes

Synthetic Polymer Fibers: Tailored Molecular Performance

Synthetic fibers are entirely human-made polymers, predominantly petroleum-based. Their primary advantage lies in the high degree of design control over physical and chemical properties.

Polyester Variants: PET, PBT, and PTT

Polyethylene terephthalate (PET) is the global standard for polyester, valued for its strength, wrinkle resistance, and shape retention. However, the PET molecular chain is rigid due to the presence of benzene rings, which can result in a "harsh" hand-feel.

Polybutylene terephthalate (PBT) and Polytrimethylene terephthalate (PTT) address these limitations. PBT has a helical main chain structure similar to nylon, giving it superior softness and elasticity. PTT (e.g., Sorona) utilizes a unique "Z-shaped" molecular conformation that provides elastic recovery significantly higher than PET, approaching the performance of spandex.

Polyamides: Nylon 6 and Nylon 6,6

Nylon is characterized by its extreme toughness, abrasion resistance, and high elasticity. Nylon 6,6 is synthesized from two monomers (adipic acid and hexamethylene diamine), creating a dense, symmetrical structure with a higher melting point (260°C) and better abrasion resistance than Nylon 6.

Nylon 6, made from a single monomer (caprolactam), has a lower melting point (220°C) but offers better dyeability and a softer hand, making it preferred for apparel weft yarns. The moisture absorption of nylon (3.5-4.5%) is higher than polyester, providing better anti-static properties but lower dimensional stability.

Technical Properties of Synthetic Fibers

Fiber Type	Tenacity (g/d)	Elongation %	Shrinkage %	Regain %	Thermal Behavior
Polyester (PET)	4.0 – 5.5	15 – 30	0.5 – 1	0.4	Melts at 256°C
Polyester (PBT)	3.0 – 4.0	30 – 40	1 – 2	0.4	Melts at 220°C
Polyester (PTT)	3.4 – 3.8	35 – 42	14 (Boil)	0.4	Melts at 228°C
Nylon 6	4.5 – 6.5	30 – 70	2 – 3	4.0	Melts at 214°C
Nylon 6,6	5.0 – 8.0	24 – 38	3 – 5	3.5	Melts at 253°C
Acrylic	2.0 – 3.5	24 – 38	1 – 2	1.5	Softens at 190°C
Polypropylene	3.5 – 5.5	15 – 45	Minimal	0.05	Melts at 165°C
Spandex	0.6 – 1.0	500 – 700	Minimal	1.0	Melts at 260°C

High-Performance Industrial Fibers

Industrial weft yarns are engineered for specific extreme conditions, such as high-impact loads, chemical exposure, or intense thermal stress.

Aramid and Specialty Polymers

Aromatic polyamides (aramids) are divided into para-aramids (Kevlar/Twaron) and meta-aramids (Nomex). Para-aramids are used for high tensile strength (up to 24 g/d) and impact resistance, while meta-aramids are used for thermal insulation and flame

resistance. Unlike standard nylons, aramids do not melt; they maintain their integrity up to 375°C before carbonizing.

Other high-performance materials include Polyphenylene sulfide (PPS), which offers exceptional chemical resistance, and Polybenzimidazole (PBI), which has the highest thermal stability of any organic fiber. Poly-phenylene benzobisoxazole (PBO), or Zylon, represents the upper limit of textile tenacity, with a breaking force of 42 g/d.

Carbon and Glass Reinforcements

Carbon fibers are categorized by their precursor: polyacrylonitrile (PAN) or petroleum pitch. PAN-based carbon fibers dominate the market due to their high tensile strength and damage tolerance, whereas pitch-based fibers are used for applications requiring extreme stiffness (high modulus) and thermal conductivity.

Glass fibers, primarily E-glass and S-glass, are the most common reinforcement materials. They offer high tensile and compressive strength, but their brittleness means they can break along crease lines during weaving.

Industrial Fiber Performance Matrix

Fiber	Tenacity (g/d)	LOI %	Shrinkage %	Regain %	Key Application
Para-Aramid	24	29	Minimal	4.5	Ballistics, Tyres
Meta-Aramid	5.5	29	Minimal	4.5	Fire-fighting gear
PPS	4.0	35	Minimal	0.6	Industrial filters
PTFE	2.0	95	Minimal	0.0	Chemical gaskets
PBI	2.6	40	Minimal	13.0	Aerospace textiles
PBO (Zylon)	42	68	Minimal	1.0	High-stress cable
Carbon (PAN)	15 – 25	N/A	Zero	0.0	Aerospace composites
E-Glass	6.0 – 9.0	N/A	Zero	0.0	Reinforcement mats

The Science of Blending: Synergistic Weft Engineering

The practice of blending different fibers in the weft allows for the optimization of fabric characteristics that cannot be achieved by a single fiber type. Blending is governed by the synergistic interaction of mechanical, thermal, and hygroscopic properties.

Standard Blend Ratios and Applications

The 65/35 Polyester/Cotton (PC) blend is the industry standard for durability and care. The polyester provides "dimensional stability"—preventing the fabric from shrinking or stretching—while the cotton provides comfort and breathability. In workwear, high-strength ratios like 70/30 are used to withstand over 100,000 abrasion cycles.

Polyester/Wool blends are critical for the suiting industry. A 55/45 or 50/50 ratio combines wool's natural drape and thermal regulation with polyester's crease recovery and wet

strength. Viscose/Polyester (PV) blends are often used as cost-effective alternatives to pure cotton or wool, offering a soft hand-feel with the structural integrity of synthetic fibers.

Physics of Blend Performance

The integration of fibers with different moisture regains affects the moisture transport through the fabric. For example, adding 10% more cotton to a PC blend typically increases moisture absorption by 2%. In three-fiber blends, such as Cotton/Polyester/Spandex, the spandex allows the weft to stretch under tension and return to its original length, preventing "bagginess" at points of stress like elbows or knees.

Blend Variant	Ratios	Tenacity	Elasticity	Shrinkage	Primary Use
PC (Standard)	65/35	High	Low	<1%	Shirts, Uniforms
PC (Comfort)	50/50	Medium	Medium	<1.5%	Sleepwear
PC (Technical)	80/20	Very High	Low	<0.5%	Industrial filter cloth
PV (Institutional)	65/35	High	Medium	<1%	School Uniforms
PV (Formal)	48/52	Medium	High	<1.5%	Suiting materials
Wool/Acrylic	50/50	Low	High	Moderate	Blankets, Sweaters
Cotton/Linen	55/45	High	Very Low	Moderate	Summer Apparel

Textile Engineering Standards and Predictive Modeling

To ensure consistency in weft yarn performance, several physical parameters are calculated using standardized textile engineering formulas.

Tenacity and Breaking Strength

The tenacity (T) of a fiber or yarn is the breaking force divided by its linear density. It is typically expressed in centinewtons per tex (cN/tex) or grams per denier (g/d):

$$T = \frac{F_{break}}{LD}$$
where F_{break} is the force at break and LD is the linear density (Tex = grams/1000m; Denier = grams/9000m).

Moisture Regain and Equilibrium

Moisture regain (R) is a measure of a fiber's hydrophilicity. It is calculated as the weight of moisture in a specimen as a percentage of its dry weight:

$$R = \frac{W_{moist} - W_{dry}}{W_{dry}} \times 100\%$$
Equilibrium moisture content is highly dependent on the relative humidity (RH) and temperature, with standard testing typically conducted at 65% RH and 20°C. Fibers with high regain, like wool and viscose, are superior for moisture management in weft yarns.

Dimensional Stability and Shrinkage

Shrinkage is the percentage change in length after a specific treatment (e.g., laundering or steam pressing). Dimensional stability is critical for weft yarns to ensure the final garment retains its shape:

$$S\% = \frac{L_{initial} - L_{final}}{L_{initial}} \times 100\%$$
Synthetic fibers like polyester are often "heat-set" to achieve dimensional stability, while natural fibers like cotton may require chemical finishes to reduce their inherent shrinkage.

Dye Affinity and Coloration Mechanics

The selection of weft yarns is often dictated by the required aesthetic finish. Dye affinity depends on the availability of chemical functional groups (chromophores) and the amorphous/crystalline ratio of the fiber.

Ionic and Non-Ionic Dyeing Systems

Cellulosic fibers (Cotton, Linen, Rayon) are primarily colored using direct or reactive dyes, which form hydrogen or covalent bonds with the hydroxyl groups of the cellulose. Protein fibers (Wool, Silk) and polyamides (Nylon) possess amino groups that interact with acid dyes through electrostatic forces.

Hydrophobic synthetics like PET polyester have no chemical sites for water-soluble dyes. Instead, they are colored using disperse dyes, which are essentially solid solutions of dye molecules within the fiber matrix, achieved by heating the fiber above its glass transition temperature (T_g) to allow the polymer chains to "open up".

Identification and Analysis Techniques

Textile engineers utilize burning tests and solubility tests for rapid fiber identification. For example, cellulosic fibers burn quickly with the smell of burning paper and leave a gray ash, while protein fibers smell like burning hair and leave a crushable black bead. Solubility tests provide definitive identification: cotton dissolves in 75% sulfuric acid, while nylon dissolves in 85% formic acid.

Implications for Weaving and Fabric Integrity

The choice of weft material directly influences the weaving efficiency and the physical characteristics of the resulting fabric. High-tenacity weft yarns are necessary for high-speed air-jet and rapier looms to prevent filling breaks. Furthermore, the elasticity of the weft yarn determines the "crimp" of the fabric—the degree to which the weft bends around the warp—which in turn affects the fabric's thickness, drape, and air permeability.

As the global textile industry shifts toward sustainability, the database of weft materials continues to expand with "green" variants like BCI cotton, recycled polyester, and bio-based polyesters like PEF. The engineering of these materials requires a careful balance between environmental impact and the rigorous technical standards of modern textile production.

Refined data from NotebookLM

Based on the technical data provided in the sources, here is the refined material properties database, categorized by fiber origin and engineering application. Values are presented within their verified ranges to align with textile engineering standards.

1. Natural Cellulosic Fibers

These fibers are characterized by their β-D-glucopyranose units and varying degrees of polymerization.

Fiber Variant	Tenacity (cN/tex)	Elongation (%)	Moisture Regain (%)	Shrinkage (%)
Upland Cotton	22 – 26	5.0 – 10.0	8.5	3 – 5
Pima Cotton	28 – 32	6.0 – 9.0	8.5	2 – 3
Egyptian Cotton	26 – 30	6.0 – 8.0	8.5	2 – 4
Organic Cotton	22 – 28	5.0 – 10.0	8.5	3 – 6
Kapok	12 – 15	1.8 – 4.2	10.0	1 – 2
Flax (Linen)	50 – 65	2.7 – 3.5	10 – 12	*Moderate*
Hemp	55 – 90	1.6	12.0	*Not Specified*
Jute	30 – 34	2.0 – 8.2	13.8	*Not Specified*
Ramie	60 – 75	3.0 – 4.0	11.0	*Not Specified*
Abaca	90 – 120	1.1	10.5	*Not Specified*
Coir	10 – 15	30.0	10.5	*Not Specified*

- **Uncertain/Qualitative Values:** Shrinkage for Flax and Cotton/Linen blends is described qualitatively as "**Moderate**" rather than by a specific percentage.

2. Natural Protein Fibers

Composed of alpha-keratin or fibroin systems, these fibers exhibit higher moisture regain and sensitivity to heat.

Fiber Type	Tenacity (g/d)	Elongation (%)	Moisture Regain (%)	Thermal Limit / Limit
Mulberry Silk	3.5 – 5.0	15 – 25	11.0	*Not Specified*
Tussah Silk	3.0 – 4.0	20 – 30	11.0	*Not Specified*
Merino Wool	1.0 – 1.7	*Not Specified*	13 – 17*	Scorch 130–140°C
Cashmere	1.0 – 1.5	*Not Specified*	13 – 15	Heat Sensitive
Mohair	1.5 – 2.0	*Not Specified*	13.0	Resilient
Vicuna	0.8 – 1.2	*Not Specified*	13.0	Very Heat Sensitive

- **Note on Moisture Regain:** While the standard testing regain for Wool is 13–17%, it is technically capable of absorbing up to **30%** of its weight without feeling damp.
- **Uncertain/Qualitative Values:** Elongation for wool and specialty hair fibers is not numerically defined in the provided tables.

3. Regenerated Fibers

These are engineered by dissolving natural polymers; they often lose strength when wet.

Fiber Variant	Tenacity (cN/tex)	Elongation (%)	Moisture Regain (%)	Shrinkage
Viscose Rayon	22 – 26	15 – 30	11 – 16	**High**
Modal	30 – 35	15 – 20	12 – 14	**Low**
Micro-modal	30 – 35	15 – 20	12 – 14	**Very Low**
Lyocell (Tencel)	38 – 44	14 – 18	11.5	**Minimal**
Acetate	11 – 13	25 – 45	6.0	**Moderate**
Triacetate	11 – 13	25 – 35	3.5	**Low**

- **Uncertain/Qualitative Values:** Shrinkage for all regenerated fibers is listed in qualitative terms (**High, Low, Minimal**) rather than numeric ranges.

4. Synthetic Polymer Fibers

Petroleum-based fibers designed for high durability and specific molecular performance.

Fiber Type	Tenacity (g/d)	Elongation (%)	Moisture Regain (%)	Thermal Behavior
Polyester (PET)	4.0 – 5.5	15 – 30	0.4	Melts at 256°C
Polyester (PBT)	3.0 – 4.0	30 – 40	0.4	Melts at 220°C
Nylon 6	4.5 – 6.5	30 – 70	4.0	Melts at 214°C
Nylon 6,6	5.0 – 8.0	24 – 38	3.5	Melts at 253°C
Acrylic	2.0 – 3.5	24 – 38	1.5	Softens at 190°C
Spandex	0.6 – 1.0	500 – 700	1.0	Melts at 260°C

- **Highlight: Nylon 6** has a notably wide elongation range (30–70%), which reflects its high elasticity and sensitivity to processing conditions.

5. High-Performance Industrial Fibers

Engineered for extreme conditions; these fibers often do not melt but carbonize at high temperatures.

Fiber Type	Tenacity (g/d)	Regain (%)	LOI %	Key Property
Para-Aramid	24	4.5	29	Ballistics/Tires
PBI	2.6	13.0	40	Aerospace textiles
PBO (Zylon)	42	1.0	68	Breaking force limit

Carbon (PAN)	15 – 25	0.0	N/A	Zero shrinkage
PTFE	2.0	0.0	95	Chemical gaskets

- **Uncertain/Qualitative Values:** Shrinkage for high-performance fibers is generically listed as "**Minimal**" or "**Zero**".

Would you like me to create a **tailored report** or a **slide deck** to present this database more formally?